

INTERFERENCE OF LIGHT



Resnick, Halliday, Krane

Q.1 What is nature of light? Explain different theories about nature of light?

NATURE OF LIGHT

Today we believe that light has a dual nature. It behaves as waves when travels such as in phenomena of interference, diffraction and polarization. It behaves as particles when collide with matter such as in phenomena photoelectric effect and Compton effect.

THEORIES OF LIGHT

The scientific study of the behavior of light is called optics. It covers reflection, refraction, diffraction, interference and polarization of light. Any successful theory about nature of light must be able to explain all these and other optical phenomena.

The earliest scientific theories about nature of light were proposed around the end of the 17th century.

HUYGENS'S THEORY OF LIGHT

Christian Huygens proposed a theory about nature of light in 1690 that light is a wave phenomenon.

NEWTON'S THEORY OF LIGHT

A rival theory to Huygens's theory was offered by Sir Isaac Newton in 1704. Newton who had discovered the visible spectrum in 1666 stated that light is composed of tiny

[M.B.D] WAVES & OSCILLATIONS For B.Sc. Classes Based on Resnick, Halliday, Particles or corpuscles emitted by luminous bodies. He was able to explain many Optical phenomena by combining this corpuscular theory with his laws of mechanics.

For more than 100 years, Newton's corpuscular theory of light was favored over the wave theory of Huygens partly because of Newton's great prestige and enough

Experimental evidence comparison between the two theories.

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YOUNG DOUBLE SLIT EXPERIMENT

Finally, some important experiments were done on diffraction and interference of light by Thomas Young and Fresnel in 1801. The results of these experiments could only be interpreted in terms of wave theory of light. The polarization of light was still another phenomenon that could only be explained by the wave theory. Thus wave theory became the dominant theory of the nature of light in the 19th century.

MAXWELL'S THEORY OF LIGHT

The Huygens wave theory of light received additional support from electromagnetic theory of light by James Clerk Maxwell. He showed electric and magnetic fields propagate together in space and their speed is identical with speed of light. It thus became clear that visible light is a form of electromagnetic radiation which constituting only a small part of the electromagnetic spectrum. Maxwell's theory was confirmed experimentally with the discovery of radio waves by Hertz in 1886.

MODERN THEORY OF LIGHT

The two general problems called ether hypothesis and explanation of photoelectric effect remained when electromagnetic theory of light is accepted.

ETHER HYPOTHESIS

A hypothetical medium was suggested by the early scientists as the carrier of light waves just as air or water carries sound waves. The ether was assumed to have some very unusual properties, e.g. being massless but having high elasticity.

A number of experiments such as Michelson and Morley experiment in 1887 were performed to give evidence of the ether but all failed to support the ether hypothesis. The

ether was shown to be unnecessary to the electromagnetic theory when special theory of relativity was published in 1905 by Einstein.

EXPLANATION OF PHOTOELECTRIC EFFECT

-The second main and serious problem was the explanation of various phenomena such as photoelectric effect that involves interaction of light with matter. The solution to the problem was again proposed by Einstein in 1905 when he explained photoelectric effect.

PHOTON THEORY OF LIGHT

Einstein extended the quantum theory of thermal radiation proposed by Max Planck in 1900 and suggested that light and other forms of electromagnetic radiations travel as tiny bundles of energy called light quanta or photons. The energy of each photon is directly proportional to its frequency.

It became apparent that light and other forms of electromagnetic radiation are emitted and absorbed during transition process when quantum theory of atomic and molecular structure is established by Bohr and other scientists.

OF LIGHT

DUAL NATURE OF LIGHT 437 Louis de Broglie introduced particle-wave duality concept in 1924. The quantum of light has successfully explained all aspects of the behavior of light.

Today we believe that light has a dual nature. It behaves as waves when travels such as in phenomena of interference, diffraction and polarization. It behaves as particles when collide with matter such as in phenomena of photoelectric effect and Compton effect.

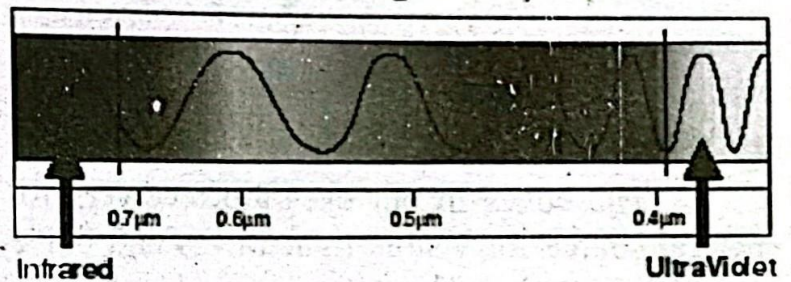
What is visible light? Explain properties of visible light and visible

VISIBLE LIGHT

The

magnetic radiation human eye and sense of sight is it or simply light. has wavelength in m to $0.7 \mu\text{m}$. It lies ed region and

Visible Light Region of the Electromagnetic Spectrum



lengths do not represent absolute limits of human vision but these are

electromagnetic visible to the human responsible for the sense called visible light or The visible light has the range of $0.4 \mu\text{m}$ to $0.7 \mu\text{m}$ between infrared ultraviolet region.

These wavelengths do not represent absolute limits of human vision but these are approximate ranges within which most people can see reasonably well under most circumstances. The people can see infrared up to at least 1050 nm while children and young adults can see ultraviolet down to about 310 nm to 313 nm under ideal conditions.

PROPERTIES OF VISIBLE LIGHT

The visible light is emitted and absorbed in tiny "packets" called photons. It exhibits properties of both waves and particles. This property of light is called Wave—particle duality. The study of light and its associated properties is called optics.

The primary properties of visible light are intensity, propagation direction, frequency and polarization. The speed of light in a vacuum having value $3 \times 10^8 \text{ m/s}$ is One of the fundamental constants of nature.

WHITE LIGHT

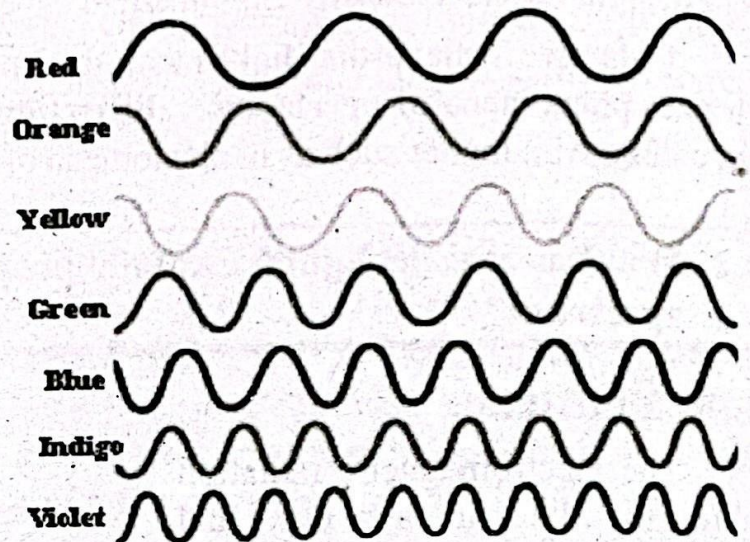
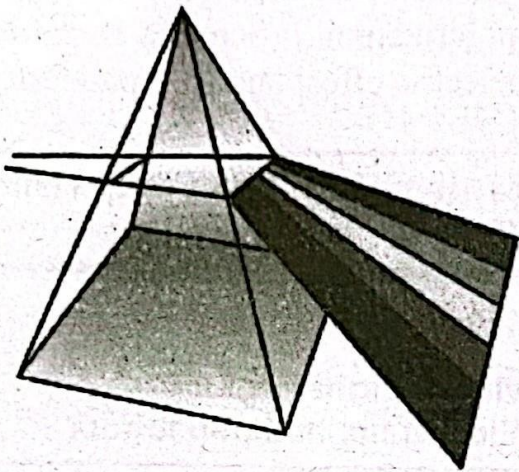
The visible light waves are the only electromagnetic waves we can see. We see these waves as the colors of the rainbow. Each color has a specific wavelength. The red color has the longest wavelength and violet color has the shortest wavelength. When all the waves are seen together, they make white light.

The white light is broken apart into colors of the visible light spectrum when white light shines through a prism. Water vapor in the atmosphere can also break apart

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Each color in a rainbow corresponds to a specific wavelength creating a rainbow. Each color in a rainbow corresponds to a specific wavelength of electromagnetic spectrum.





HOW DO WE "SEE" USING VISIBLE LIGHT?

The cones in our eyes are receivers for tiny visible light waves. The Sun is a natural source for visible light waves and our eyes see the reflection of sunlight off the objects around us. The color of an object that we see is the color of light reflected. All other colors are absorbed. The light bulbs are another source of visible light waves.

WHAT DOES VISIBLE LIGHT SHOW US

It is true that we are blind to many wavelengths of light. The instruments that can detect different wavelengths of light are used to study the Earth and the Universe. Many instruments that detect visible light can see farther and more clearly than our eyes. That is why we use satellites to look at the Earth and telescopes to look at the sky. more clearly is to look at the sky.

VISIBLE SPECTRUM

The portion of electromagnetic spectrum that is visible to the human eye is called visible spectrum. The electromagnetic radiation in this range of wavelengths is called visible light or simply light. A typical human eye will respond to wavelengths from about 0.4 μm to 0.7 μm .

This spectrum does not contain all the colors that human eyes and brain can distinguish. The unsaturated colors such as pink or purple variations such as magenta are absent because they can be made only by a mix of multiple wavelengths. The colors containing only one wavelength are called pure colors.

OPTICAL WINDOW

The region of the electromagnetic spectrum that allows wavelengths to pass largely un—attenuated through the Earth's atmosphere is called optical window. For

HT example midday sky appears blue because clean air scatters blue light more than red light. The optical window is also called visible window because it overlaps the human

The near infrared window lies just out of the human vision. The medium wavelength infrared window and long wavelength infrared window can be experienced

h_v ani

Many species can see light with frequencies outside the "visible spectrum". The bees and many other insects can detect ultraviolet light which helps them to find nectar in flowers. The plant species that depend on 'insect pollination may owe reproductive success to their appearance in ultraviolet light rather than how colorful they appear to humans. The birds too can see into the ultraviolet (300—400 nm).

SPECTRAL COLORS

The colors that can be produced by visible light of a narrow band of wavelengths (monochromatic light) are called pure spectral colors. The various color ranges indicated in the table below are an approximation: The spectrum is continuous with no clear boundaries between one color and the next.

COLOR	FREQUENCY	WAVELENGTH
Violet	668-789 THz	380-450 nm
Blue	606-668 THz	450-495 nm
Green	526-606 THz	495-570 nm

Yellow	508-526 THz	570-590 nm
Orange	484-508 THz	590-620 nm
Red	400-484 THz	620-750 nm

. What is interference of light? Explain constructive and destructive Interference?

INTERFERENCE OF LIGHT

The resultant Of two identical light waves having same frequency, same Wavelength and same amplitude when they are travelling in the same direction is called interference of light. The resultant waveform depends upon phase difference or path difference of two light waves.

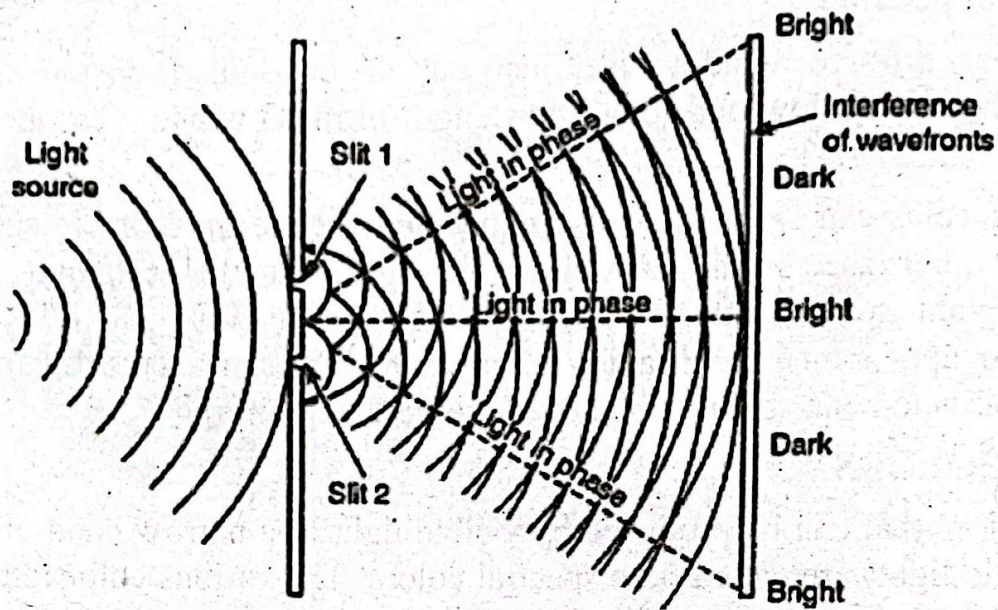
BRIGHT FRINGES

The amplitude and intensity of resultant waveform becomes maximum when two

Identical light waves combine together in phase. The two light waves are called in phase

when they have phase difference of integral multiple of 2π . The locations of maximum intensity are called maxima or bright fringes.



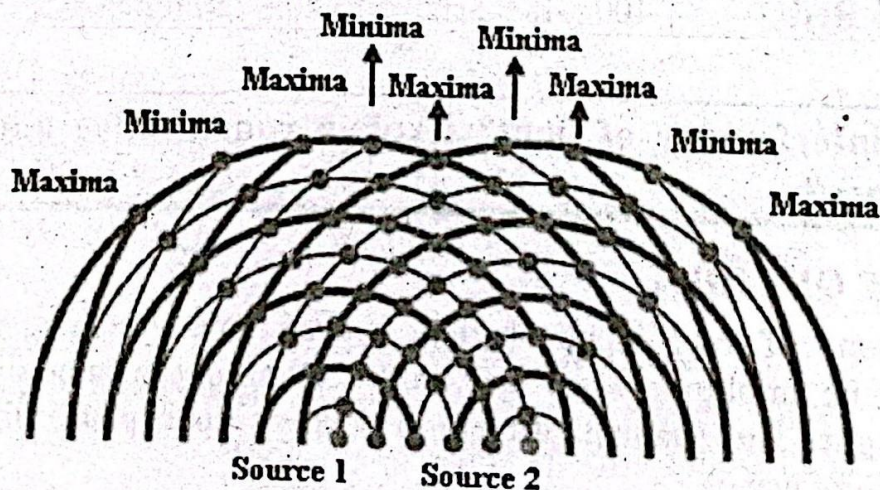


DARK FRINGES

The amplitude and intensity of resultant waveform becomes minimum when two identical light waves combine together out of phase. The two light waves are called out of phase when they have phase difference of odd multiple of π . The locations of minimum intensity are called **minima** or **dark fringes**.

INTERFERENCE PATTERN

The pattern of maxima and minima formed when two identical light waves combine together is called interference pattern as shown in fig. The bright fringes and dark fringes are equally spaced.



CONDITIONS OF INTERFERENCE

The two light waves must be monochromatic and phase coherent for detectable interference.

DARK FRINGES occur when two of minimum

The pattern of maxima and minima

light waves

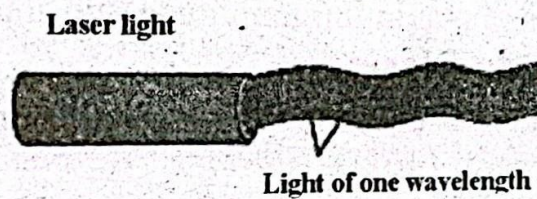
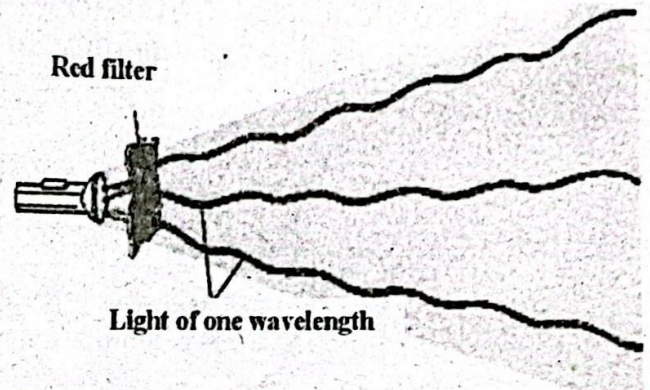
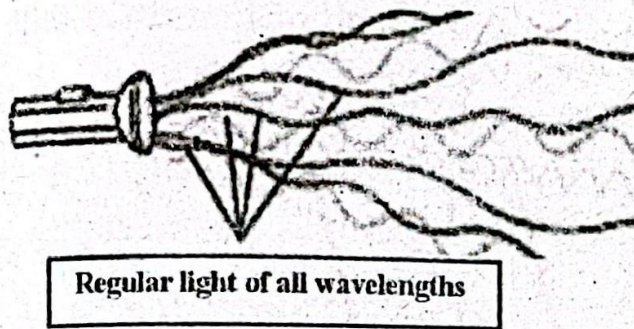
The light such as laser having single wavelength and one colour is called
MONOCHROMATIC LIGHT.

monochromatic light.

The white light contains many wavelengths of light. A filter transmits light of one wavelength or monochromatic light but weakens the original beam. The lasers produce

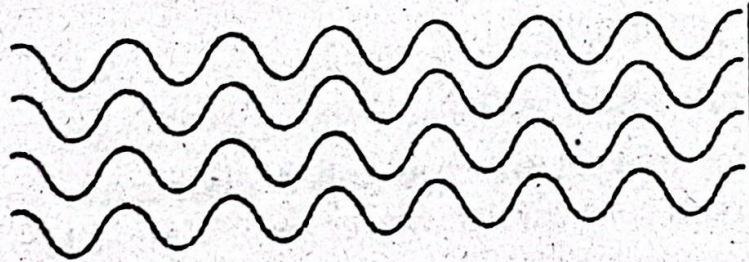


light that is monochromatic and coherent. weakens the original beam. The lasers produce



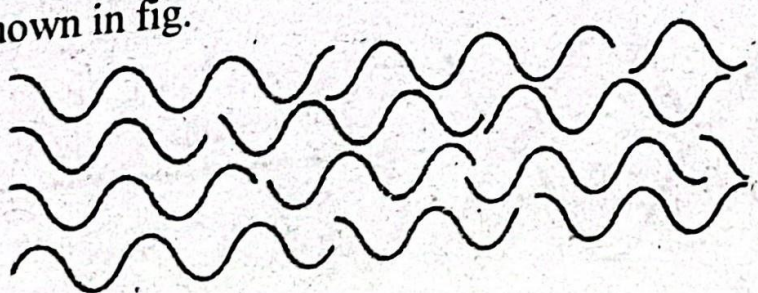
PHASE COHERENT

The two waves are called phase coherent when they have fixed phase relationship between them as shown in fig. Monochromatic light sources, which emit waves of constant phase difference or send out crest or trough at the same time are called sources.



INCOHERENCE

The two waves are called incoherent when phases of each wave change randomly at random times as shown in fig.



The two waves are called phase coherent when they have fixed phase relationship between

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 them as shown in fig. The monochromatic light sources which emit waves of constant phase difference or send out crest or trough at the same time are called coherent sources.

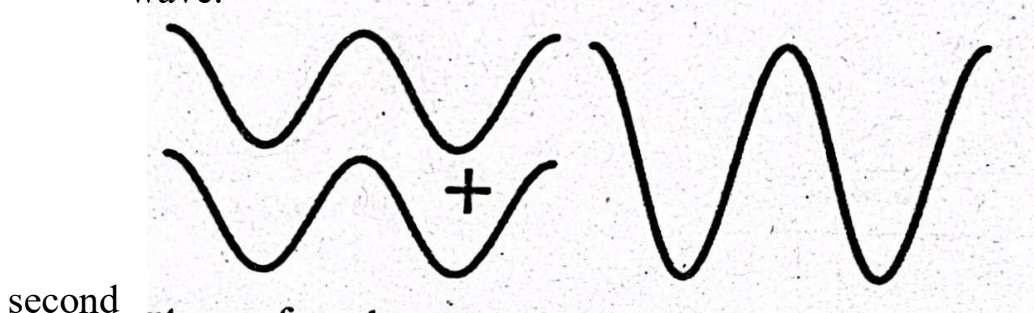
INCOHERENCE

TYPES OF INTERFERENCE

The interference of light has two types called constructive interference and destructive interference.

CONSTRUCTIVE INTERFERENCE

The interference of light waves is called constructive interference when two monochromatic and phase coherent light waves combine together in phase. The two light waves are called in phase when they have phase difference of integral multiple of 2π . It means crest of first wave falls on crest of second wave and trough of first wave falls on trough of second wave.



The resultant waveform has maximum amplitude and intensity. The locations of maximum intensity are called bright fringes.

The condition of constructive interference or bright fringes in terms of phase difference is given as

$$\text{Phase difference} = m(2\pi)$$

Where $m = 0, 1, 2, 3, 4, 5$

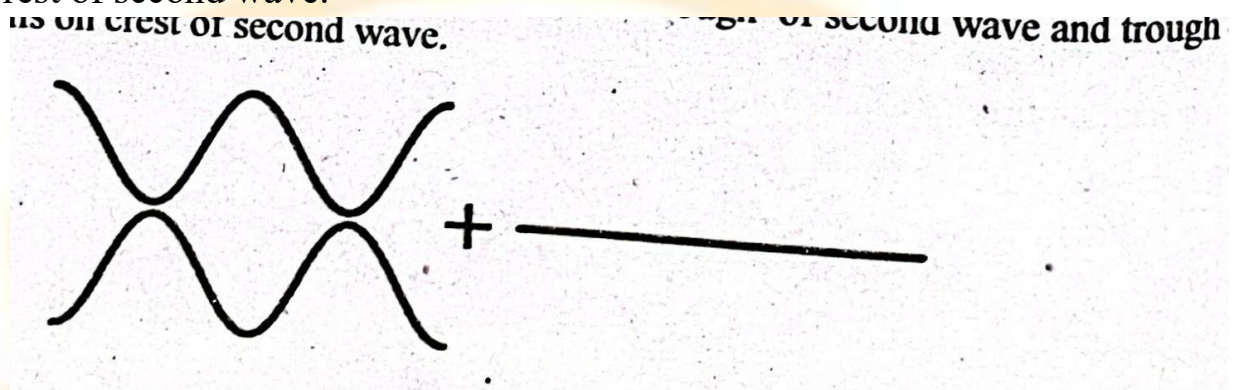
The condition of constructive interference or bright fringes in terms of path difference is

Path difference = $m\lambda$

It means bright fringes are formed when path difference between two light waves is integral multiple of wavelength.

DESTRUCTIVE INTERFERENCE

The interference of light waves is called destructive interference when two monochromatic and phase coherent light waves combine together out of phase by 180° . The two light waves are called out of phase by 180° when they have phase difference of odd multiple of π . It means crest of first wave falls on trough of second wave and trough of first wave falls on crest of second wave.



The resultant waveform has minimum amplitude and intensity. The locations of minimum intensity are called dark fringes.

The condition of destructive interference or dark fringes in terms of difference is given as

$$\text{Phase difference} = \left(m + \frac{1}{2}\right)(2\pi)$$

Where $m = 0, 1, 2, 3, 4, \dots$

The condition of destructive interference or dark fringes in terms of path difference is given as

$$\text{Path difference} = \left(m + \frac{1}{2}\right)\lambda$$

It means dark fringes are formed when path difference between two light waves is odd multiple of halfwavelength.

RELATION BETWEEN PATH DIFFERENCE AND PHASE DIFFERENCE

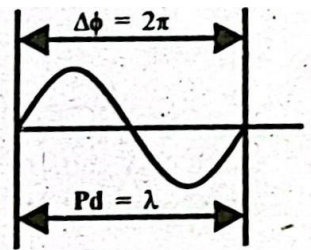
When path difference = X , Phase difference = $2\pi \frac{X}{\lambda}$

When path difference = λ , Phase difference $\Delta\phi = \frac{2\pi}{\lambda}$

When path difference = d , Phase difference $\Delta\phi = \frac{2\pi}{\lambda} d$

Hence phase difference $\Delta\phi$ and path difference d are related as

$$\Delta\phi = \frac{2\pi}{\lambda} d$$



Q.4 Describe Young double slit experiment in detail? (PU. 2009, SU. 2012)

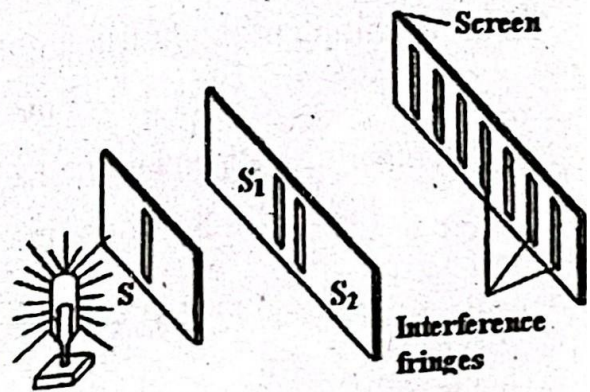
YOUNG DOUBLE SLIT EXPERIMENT

The phenomenon of interference of light was experimentally confirmed by Young through double slit experiment in 1801. This experiment gave a very strong support to the wave theory of light.

EXPERIMENTAL ARRANGEMENT

The experimental arrangement of Young's double slit experiment consists of a monochromatic light source S placed in front of narrow slits S_1 and S_2 having separation d . A screen used to observe interference pattern is placed at distance L from slits as shown in fig.

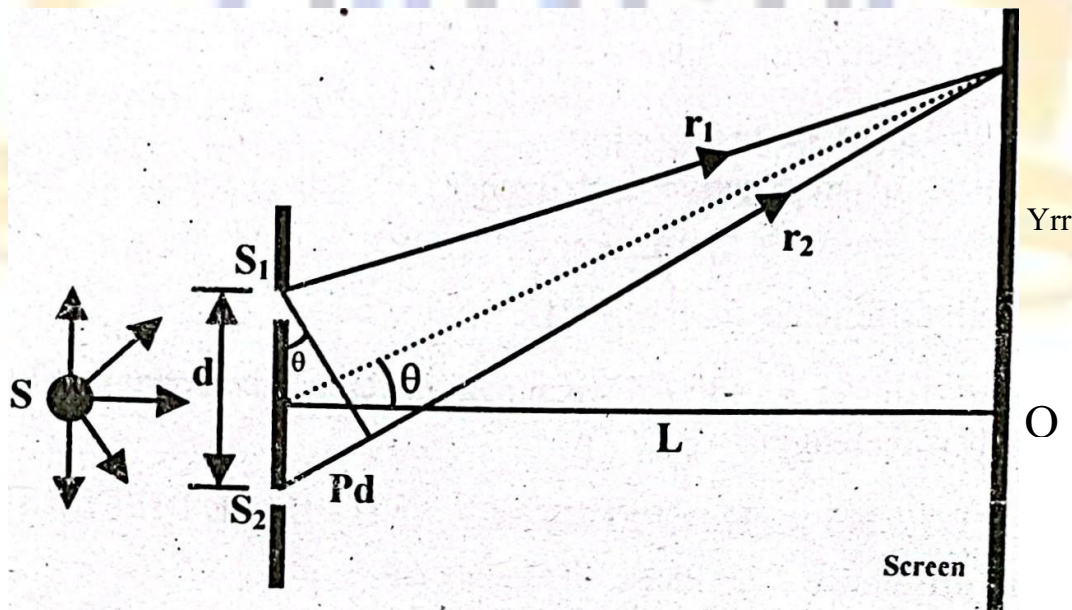
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WORKING

The light is emitted from source. It emerges from the pinhole S and spreads as spherical wavefronts in front of slits S_1 and S_2 . Now light waves emerging through S_1 and S_2 reach at P after covering distance r_1 and r_2 . The both light rays have path difference

when meet at screen.



S,

The formation of bright fringes and dark fringes called maxima and minima respectively depends upon path difference $Pd = d \sin \theta$ between two light rays when they reach at

point P on screen. The interference pattern is shown in fig.

The path difference of two rays in diagram of Young's double slit experiment is

$$Pd \equiv d \sin \theta$$

bright fringes or constructive interference is

$$Pd = m\lambda$$

The condition of fringes or constructive

On comparing both equations

$$d \sin \theta = m\lambda$$

$$\sin \theta = \frac{m\lambda}{d}$$

This is called condition of bright fringes in Young double slit experiment

----- (1)

CONDITION OF DARK FRINGES

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The path difference of two rays in diagram of Young's double slit experiment is

$$Pd \equiv d \sin \theta$$

The condition of dark fringes or destructive interference is

$$Pd = \left(m + \frac{1}{2}\right) \lambda$$

On comparing both equations

$$d \sin \theta = \left(m + \frac{1}{2}\right) \lambda$$

$$\sin \theta = \left(m + \frac{1}{2}\right) \frac{\lambda}{d}$$

(2)

This is called condition of dark fringes in Young double slit experiment

POSITION OF BRIGHT FRINGES ON SCREEN

From diagram of Young double slit experiment

$$\tan \theta = \frac{y_m}{L}$$

For small angles, $\sin \theta \approx \tan \theta$

$$\sin \theta = \frac{y_m}{L}$$

(3)

By eq (1) and eq (3)

$$\frac{m \lambda}{d} = \frac{y_m}{L}$$

Comparing eq (1) $m \lambda$

This is called position of bright fringes on screen. The position of zero order bright fringe is at distance $y_0 = 0$ for $m = 0$. The first bright fringe is at distance $= \lambda L/d$ $m = 1$ and so on.

POSITION OF DARK FRINGES ON SCREEN

From diagram of Young double slit experiment

$$\tan \theta =$$

For small angles, $\sin \theta \approx \tan \theta$ (4) $\sin \theta$

Comparing eq (2) and eq (3)

$$\frac{y_m}{L} = \left(m + \frac{1}{2}\right) \frac{\lambda}{d}$$

$$y_m = \left(m + \frac{1}{2}\right) \frac{\lambda L}{d}$$

This is called position of dark fringes on screen. The position of first dark fringe is at distance $y_0 = \lambda L/2d$ for $m = 0$ and so on.

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FRINGE WIDTH

The separation between two consecutive bright fringes or consecutive dark fringes is called fringe width or fringe separation and denoted by Δy . It is also called linear separation of fringes.

$$\Delta y = y_{m+1} - y_m = (m+1) \frac{\lambda L}{d} - \frac{m\lambda}{d}$$

$$\Delta y = \frac{\lambda L}{d} \quad \text{----- (5)}$$

The fringe width increases when separation L between slits and screen increases. but fringe spacing decreases when slit separation d of slits increases. The fringe width increases when wavelength of light increases.

ANGULAR SEPARATION

The angular difference $\Delta \theta$ between two consecutive bright fringes or two consecutive dark fringes is called angular separation.

$$\Delta y = y_{m+1} - y_m = L \sin \theta_{m+1} - L \sin \theta_m$$

$$\Delta y = L(\sin \theta_{m+1} - \sin \theta_m)$$

For small angles, $\sin \theta \approx \theta$

$$\Delta y = L(\theta_{m+1} - \theta_m) = L \Delta \theta$$

$$\Delta \theta = \frac{\Delta y}{L} \quad \text{----- (6)}$$

Comparing eq (5) and eq (6).

$$\Delta \theta = \frac{\Delta y}{L}$$

$$\Delta \theta = \frac{\lambda}{d}$$

This is called formula for angular separation of fringes.

ORDER OF BRIGHT AND DARK FRINGES

The order of bright and dark fringes on screen is given in table. The values of m and order of bright fringes is same. The values of m and order of dark fringes is not same.

Q.5 Explain the analytical treatment of intensity in Young double slit experiment and describe conditions of maximum and minimum intensity by using trigonometric method and phasor method?
(PU. 2003, 2006, 2011, 2012, 2013, SU. 2011, GCUF. 2012, 2013).

Value of m	Position of bright fringes	Position of dark fringes	Order of bright fringes	Order of dark fringes
	$y_0 = 0$	$y_0 = \frac{\lambda L}{2d}$		fringes
$m=1$		$y_1 = \frac{3\lambda L}{2d}$	Zero order	1st order
			1st order	2 nd order
	$\frac{21}{L}d$	$y_2 = \frac{5\lambda L}{2d}$	2 nd order	3 rd order

	$y_3 = \frac{3\lambda L}{d}$	$y_3 = \frac{7\lambda L}{2d}$	3 rd order	4th order
so on	so on	so on	so on	so on

INTENSITY OF LIGHT

The term intensity is used to describe the rate at which light spreads over a surface of a given area having some distance from a source. The intensity is inversely proportional to the distance from the source and directly proportional to power of the source. The rate at which light energy is emitted by the source is called power of light.

The power transferred per unit area is equal to intensity of light. Its SI unit is called watt/m².

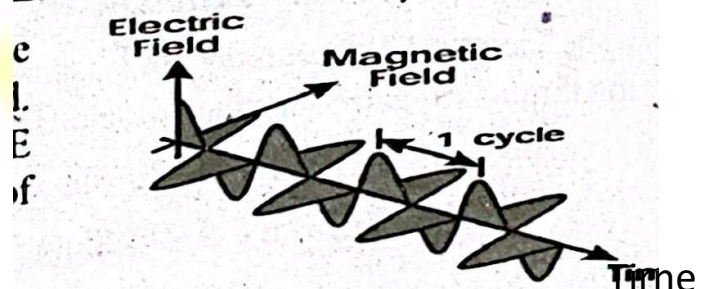
$$I = \frac{P}{A}$$

ANALYTICAL TREATMENT OF INTENSITY OF LIGHT

The light is a combination of time varying electric field and magnetic field.

We generally consider electric field E Component of light because effects of magnetic field on detectors is very small.

INTENSITY OF LIGHT





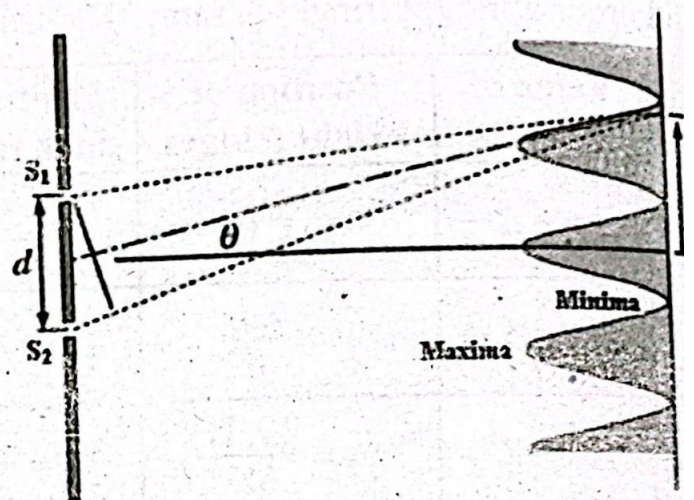
Now consider a monochromatic light source S of wavelength λ placed in front of two narrow slits S_1 and S_2 as shown in diagram.

The electric field components E_1 and E_2 of two waves emerging from S_1 and S_2 having phase difference ϕ is given as

$$E_1 = E_0 \sin \omega t$$

$$E_2 = E_0 \sin(\omega t + \phi)$$

Where E_0 is amplitude and ω is angular frequency of each time varying component of electric field because both light waves are identical.



ADDITION OF WAVES BY TRIGONOMETRIC METHOD

The resultant electric field due to combination of two identical light waves is

$$E = E_1 + E_2 = E_0 \sin \omega t + E_0 \sin(\omega t + \phi)$$

$$E = E_0 [\sin \omega t + \sin(\omega t + \phi)]$$

$$E = 2 E_0 \sin \left[\frac{\omega t + \omega t + \phi}{2} \right] \cos \left[\frac{\omega t - \omega t - \phi}{2} \right]$$

$$E = 2 E_0 \sin \left(\omega t + \frac{\phi}{2} \right) \cos \left(\frac{\phi}{2} \right) \quad [\text{Use } \cos(-\theta) = \cos \theta]$$

$$E = 2 E_0 \cos \beta \sin(\omega t + \beta)$$

$$E = E_0 \sin(\omega t + \beta)$$

Where E_0 is called amplitude of resultant electric field written as

$$2 E_0 \cos \beta \quad \text{-----} \quad (1)$$

The amplitude E_0 of resultant electric field of two electric fields E_1 and E_2 can also be obtained by phasor method as

ADDITION OF WAVES BY PHASOR METHOD

The electric field components E_1 and E_2 of two waves reaching at screen after emerging through S_1 and S_2 having phase difference are

$$E_1 = E_0 \sin \omega t$$

$$E_2 = E_0 \sin(\omega t + \phi)$$

PHAS

DIAGRAM OF FIRST COMPONENT OF ELECTRIC FIELD

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The phasor diagram of first electric field E_1 is shown in fig. The magnitude of amplitude E_0 is represented by phasor vector which rotates counterclockwise direction with angular

frequency ω .

The alternating electric field E_1 is represented by the vertical projection of this phasor on the vertical axis.

PHASOR DIAGRAM OF SECOND COMPONENT OF ELECTRIC FIELD

The phasor diagram of second electric field E_2 is shown in fig. It has same amplitude E_0 but phase difference with respect to E_1 . The magnitude of amplitude E_0 is represented by phasor vector which makes angle with first vector. The alternating electric field E_2 is represented by the projection of this phasor on the vertical axis.

The sum of E_1 and E_2 is equal to E which is sum of the projections of the two phasors on the vertical axis.

PHASOR DIAGRAM OF RESULTANT ELECTRIC FIELD

The resultant electric field E of component electric field E_1 and E_2 can be obtained by vector addition rule as

Place the first phasor vector of magnitude E_0 with second phasor vector. of magnitude E_0

second be joined with head of first. Now joining the tail of first phasor vector with head of second phasor vector. It gives the resultant in such a way that tail of

head resultant phasor vector of magnitude E_0 as shown in phasor diagram.

The projection of phasor vector E_0 on vertical axis is equal to $E_0 \sin(\omega t + \phi)$

This is the resultant electric field reaches at screen. The triangle having sides E_0 , E_0 and E_0 isosceles because its two sides are equal. Hence two interior angles of triangle are also equal as shown in fig,

From triangle

$$\begin{aligned} \beta + \beta + (180^\circ - \phi) &= 180^\circ \\ \beta &= \frac{\phi}{2} \end{aligned}$$

The right angled triangle with hypotenuse $\frac{E_0}{2}$ gives

$$E_n = 2E_0 \cos \beta$$

This is the amplitude of resultant wave. It is same as in eq(I).

INTENSITY OF RESULTANT ELECTRIC FIELD.

The amplitude of resultant wave derived in eq(I) can be written as

$$E_n = 2E_0 \cos \beta$$

Squaring on both sides

$$\frac{I_{\theta}}{I_0} = 4 \cos^2 \beta \quad \text{----- (2)}$$

The intensity of light is directly proportional to square of amplitude. The intensity of individual light wave having amplitude E_0 is given as

$$I_0 = K E_0^2$$

The intensity of resultant light wave having amplitude E_{θ} is given as

$$I_{\theta} = K E_{\theta}^2$$

eq

$$\frac{E_{\theta}^2}{E_0^2} = \frac{I_{\theta}}{I_0} \quad \text{----- (3)}$$

eq (2) and eq (3)

$$\frac{I_{\theta}}{I_0} = 4 \cos^2 \beta$$

$$I_{\theta} = 4 I_0 \cos^2 \beta$$

$$I_{\theta} = 4 I_0 \cos^2 \frac{\phi}{2}$$

$$I_{\theta} = 4 I_0 \cos^2 \frac{\phi}{2}$$

Divide both eq

Comparing eq (2) and eq

The intensity of resultant waveform depends upon $\cos^2 \theta$

CONDITION OF MAXIMUM INTENSITY

The intensity will be maximum when $\cos$ is max. The maximum value of $\cos^2$ is ± 1 . This is possible when

$$2$$

$$0, n, \dots, 27t, 37t$$

$$\psi = 0, 1(2\pi t), 2(2\pi t), 3(2\pi t) - - 4) = m(2\pi t)$$

CONDITION OF MINIMUM INTENSITY

The intensity will be minimum when $\cos \phi$ is minimum. The minimum value of

$\cos$

is possible when

2 is

$$\frac{\phi}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} - - - - -$$

$$\phi = 1(\pi), 3(\pi), 5(\pi), 7(\pi) - - - - -$$

$$\phi = \left(m + \frac{1}{2}\right)(2\pi)$$

The condition of minimum intensity corresponding to dark fringes or destructive of path difference is

$$Pd = \left(m + \frac{1}{2}\right)\lambda$$

Intensity is given as

$$I_{\theta} = 4 I_0 (0)^2 = 0$$

zero. This is possible when

interference in terms of

The condition of maximum intensity corresponding to bright fringes or constructive interference in terms of path difference is

$$Pd = m\lambda$$

The maximum intensity is given as

$$I_{\theta} = 4 I_0 (\pm 1)^2 = 4 I_0$$

CONDITION OF MINIMUM INTENSITY

The minimum intensity is given as

$$I_{\theta} = 0$$

Q.6 Explain interference in thin films? (PU. 2014, sU. 2014)

THIN FILMS

A layer of a material having thickness in the nanometer range is called thin film. The thickness of thin film is comparable with wavelength of light.

EXAMPLES

O The surface of a soap bubbles is called thin film. The beautiful color formation in soap bubbles is due to interference in thin films.

The surface of oil on water is called thin film. The color formation in oil film on the surface of water is due to interference in thin film.

APPLICATIONS

O The household mirror is a familiar application of a thin film. A thin metal coating made on the back of a sheet of glass to form a reflective interface.

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[M.B.D] WAVES & OSCILLATIONS For B.Sc. Classes Based on Resnick, Halliday, Krane

② Thin films are used to produce thin film batteries. Thin films are also used in dye sensitized solar cells.

O Ceramic thin films are widely used in technology. •It is used for protection of substrate materials against corrosion, oxidation and wear. The use of such coatings on cutting tools extends their life.

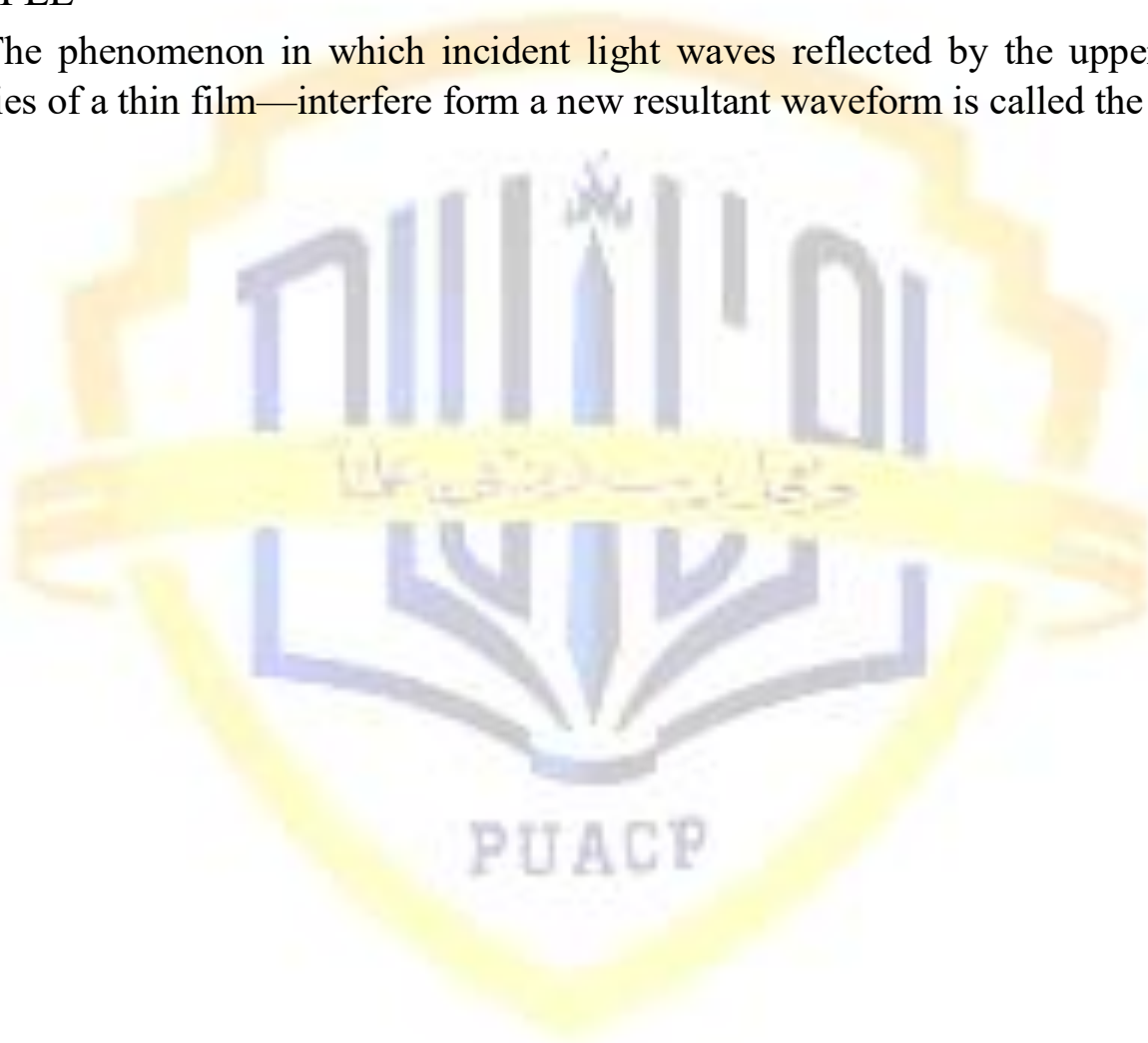
- The thin film is useful for reflective, anti-reflective coatings or self-cleaning glass. It is also useful to separate layers of insulators, semiconductors and conductors form integrated circuits.

THIN FILM INTERFERENCE

The phenomenon in which incident light waves reflected by the upper and lower boundaries of a thin film interfere with one another to form a new wave is called thin film interference.

PRINCIPLE

The phenomenon in which incident light waves reflected by the upper and lower boundaries of a thin film—interfere form a new resultant waveform is called the basic



Boundaries of a thin film interfere with a new resultant wavefront is called the basic principle of thin films.

TYPES OF THIN FILM

The thin films have two types.

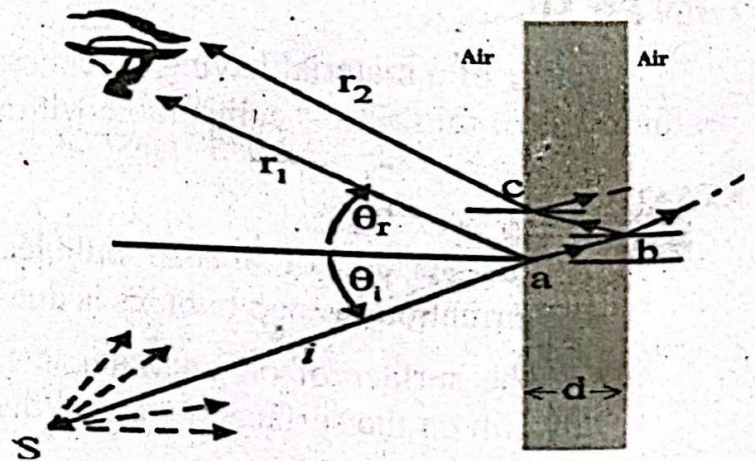
- ① Thin film of uniform thickness
- ② Thin film of non-uniform thickness.

WORKING OF THIN FILM OF UNIFORM THICKNESS

Consider a transparent film of uniform thickness d . It is illuminated by monochromatic light having wavelength λ from a point source S .

The incident ray I from the source S falls on film at "a" and makes angle θ_i with normal line. It is reflected and enters the eye as ray r_1 .

The incident ray also enters the film at "a". It is again reflected at "b". This reflected ray leaves the film at "c" and enters the eye as ray r_2 .



The rays r_1 and r_2 are parallel and phase coherent. Both rays have phase difference.

The formation of bright fringes and dark fringes depends upon phase difference.

difference. The bright fringes are formed when phase difference is $m\lambda$. The dark fringes are formed when phase difference is $m\lambda + \frac{\lambda}{2}$.

The dark

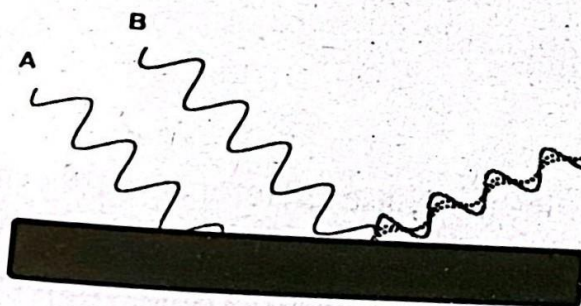
PHASE INTERACTION

[hich

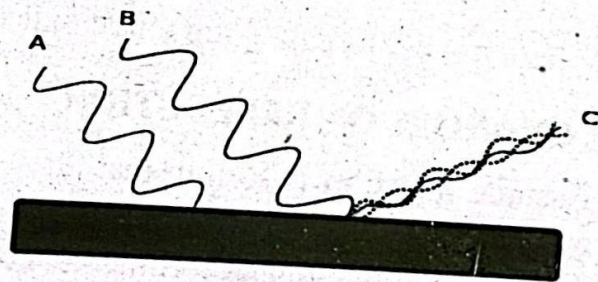
Consider two incident light beams A and B produce a reflected beam shown by dashed waves. The beam A takes reflection from lower and B takes reflection from upper surface of the thin film. These reflected beams combine to produce a resultant beam C. The resultant beam is relatively strong if reflected beams are in phase as shown in first fig. The resulting beam is weak if reflected have

The phase relationship of two reflected beams depends on the relationship between wavelength of light beams and thickness of the film. The two reflected will tend to reinforce with one another if total distance travelled by beam A inside the film is an integral multiple of the wavelength of the beam as shown in first fig.

The two reflected beams will tend to cancel with one another if total distance traveled by beam A inside the film is odd multiple of the wavelength of the beam as shown in second fig.



First figure



Second figure

NORMAL INCIDENCE

The light ray is incident normally when $\theta_i = 0$. The portion of light ray is reflected from boundary of film at position a while portion of light ray enters into thin film and reaches at back boundary of film after covering distance d . It is reflected back and emerges into same medium after covering distance d .

The geometrical path difference of two rays is close to $2d$. The resultant wave reflected from the film will interfere maximum if distance $2d$ is integral multiple of Wavelength of light. This statement must be modified for two reasons.

DIFFERENCES IN OPTICAL PATH LENGTH

The Wavelength of light in thin film must be considered rather than wavelength in air. It means we are concerned with optical path lengths rather than geometrical path-length. The speed of light and its wavelength reduces by the same factor when light

from one medium to another medium but frequency of light is unchanged.

Cams

Halliday, Krane

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The speed of light and its wavelength in first medium are related as

$$f = \frac{V_1}{\lambda_1}$$

The speed of light and its wavelength in second medium are related as

On comparing both equations

$$\lambda_2 = \lambda_1 \frac{V_2}{V_1}$$

Assume that wavelength of light in air is λ_1 having speed $V_1 = c$ while wavelength λ_2 in medium of film

$$\frac{c}{\lambda_1} = \frac{V_2}{\lambda_2}$$

of light is $\lambda_2 = \lambda_1 \frac{V_2}{c}$ in medium of film having velocity $V_2 = V$.

n

Where $n = c/V$ is called refractive index of film.

PHASE CHANGES ON REFLECTION

Assume that film is so thin that $2d < \lambda$. It means phase difference between two light waves would be close to zero when light falls on it normally. Such a film must

appear bright on reflection according to condition of constructive interference but it appears dark experimentally.

The film appears dark due to phase change of 180° when it is reflected at the air—film interface. The portion of light ray undergoes a phase change of 180° when it is reflected at front surface of film. The portion of ray transmitted in film does not change phase. This transmitted part is again reflected from back boundary of film and emerges in the same medium air but does not change phase. Now two rays combine together out of phase at the air—film interface and produces destructive interference as shown in fig.

INTENSITY CONDITIONS

The maximum intensity of resultant waveform is given as

$$I_{\text{max}} = I_1 + I_2$$

Now apply conditions of difference in optical path length and phase changes on reflection for normal incidence.

$$2d = m\lambda + \frac{\lambda}{2}$$

The term $\frac{\lambda}{2}$ is introduced because upon reflection there is a phase change of 180° or path difference of half wavelength.

Put $d = X/n$ in above eq

$$2d = \left(m + \frac{1}{2}\right) \frac{\lambda}{n}$$

$$2nd = \left(m + \frac{1}{2}\right) \lambda$$

Where $m = 0, 1, 2, 3, \dots$. This is condition of maximum intensity for light on thin film.

normal incidence of light on

The minimum intensity of resultant waveform is given as

$$I_{\text{min}} = 0$$

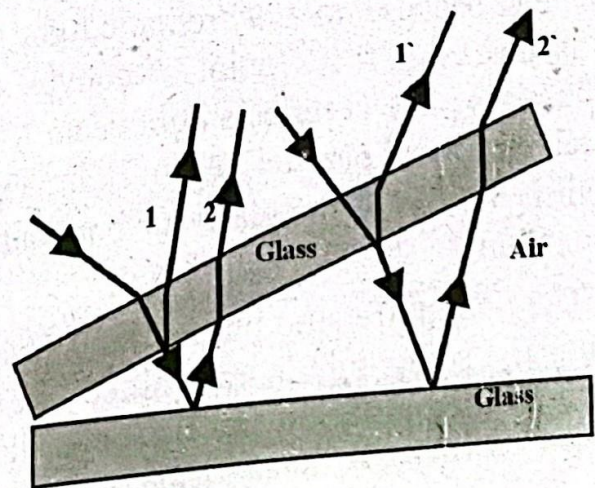
It shows condition of maximum intensity is replaced with condition of minimum intensity when light is incident normally on thin film. Hence condition of minimum intensity is

$$2nd \equiv m\lambda$$

These equations hold when index of refraction of the film is either greater or less than the indices of the media on each side of the film.

WORKING OF THIN FILM OF NON-UNIFORM THICKNESS

The constructive interference occurs in certain parts of film while destructive interference in other parts of film when thickness of the film is non-uniform or wedge shaped.



interference in other parts of film when thickness of the film is non-uniform or wedge shaped.

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[M.B.D] WAVES & OSCILLATIONS For B.Sc. Classes Based on Resonance, Monday, K. Rane

The bands of maximum and minimum intensity appear called fringes of Constant thickness. The width and spacing of fringes depends on variation of thickness of the film.

The light reflected from various parts of the film is modified by the various constructive or destructive interferences when film is illuminated with white light rather than monochromatic light. It explains brilliant colors of soap bubbles and oil slicks.

Q.7 Write a brief note on Newton rings? (PU. 2003, 2004, 2011, 2012, GCUF. 2013)

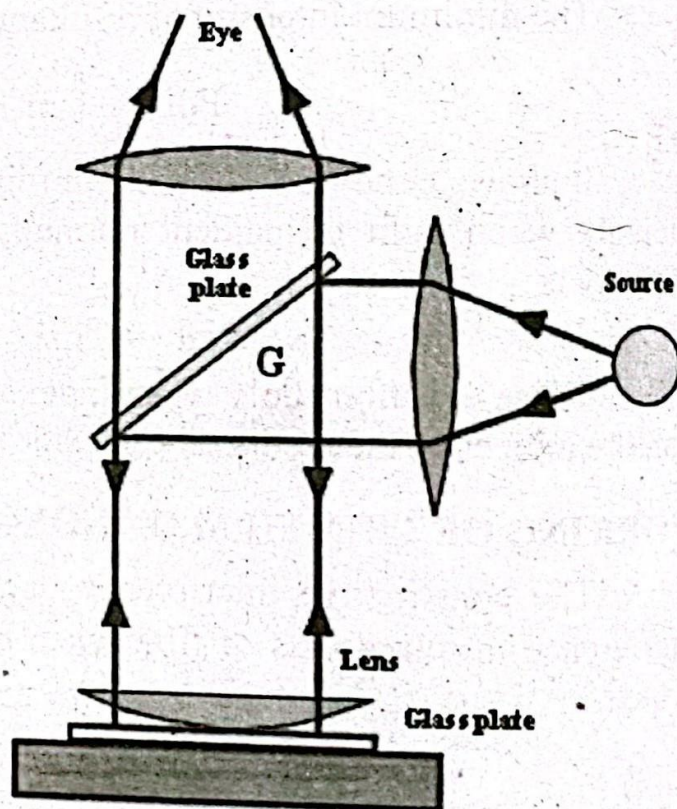
NEWTON'S RINGS

The phenomenon in which interference pattern is created due to reflection of light

between a spherical surface and an adjacent flat surface is called Newton rings. The interference pattern consists of a series of alternating -bright and dark concentric circles which are formed when a plano convex lens placed on a glass plate is illuminated by monochromatic light.

EXPERIMENTAL ARRANGEMENT

Consider a plano convex lens of large focal length placed on a plane glass plate. The spherical glass is placed on plane glass plate in such a way that convex side of spherical glass touches the flat glass plate. A monochromatic light source is placed at focus point of convex lens. A glass plate G is used to reflect light rays towards plano convex lens. A microscope is used to observe Newton rings.



WORKING

A parallel beam of light falls on glass plate G because monochromatic light source is placed at focus point of convex lens. The glass plate G reflects light beam towards plano convex lens. Now plano convex lens receives normal parallel rays of light. These light rays pass through the plano convex lens until it reaches to the glass—air boundary. It means light goes from denser medium

The light suffers no phase change when it passes through this boundary. Also at boundary, some light is transmitted into the air and some light is reflected. The light that is transmitted to the air travels some distance before it is reflected at the flat surface below. The air—glass boundary causes a half cycle phase shift because the lower

REFLECTION OF LIGHT

refractive index than the glass. The two reflected rays now travel in the same direction. The resultant is interference.

INTERFERENCE PATTERN

The bright rings are caused by constructive interference between the light rays

Reflected from both surfaces while dark rings are caused by destructive interference. The outer rings are closely spaced than inner rings because slope of the convex lens surface increases outwards.

CENTRAL SPOT OF NEWTON RINGS

The thickness of air film is Zero at contact point between plano convex lens and plane glass plate. The zero thickness of air film corresponds to bright fringe but central spot of Newton rings obtained experimentally is dark as shown in fig.

WHY CENTRAL SPOT OF NEWTON RINGS IS DARK

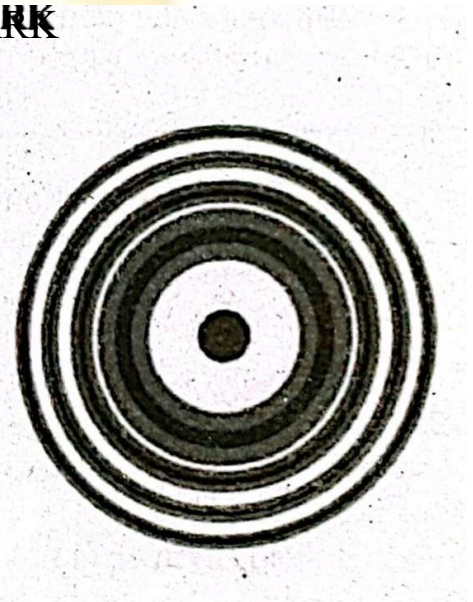
The contact point between plano convex lens and plane glass plate behaves as denser medium. The portion of light reflected at this point suffers a phase change of 180° . The portion of light transmitted through it does not change its phase and emerges in the same medium.

Now light reflected from top surface and bottom surface when combine together has phase change of 180° . It is condition of dark fringes.

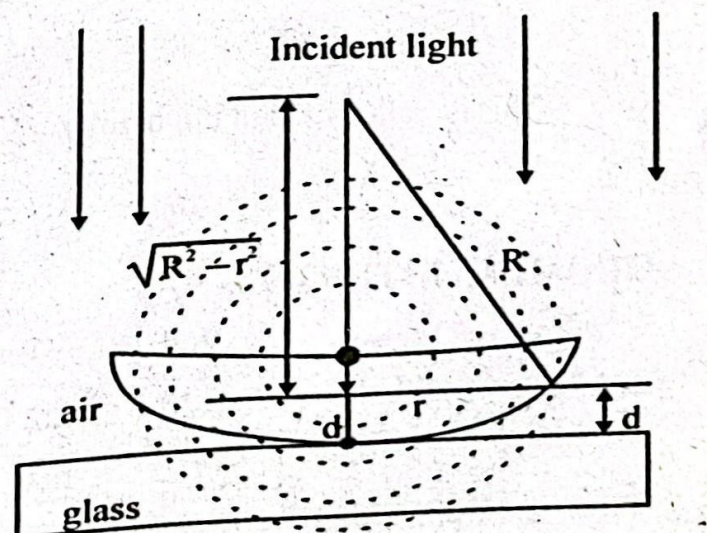
Hence central spot of Newton rings appears dark on reflection.

MATHEMATICAL TREATMENT

Consider a plano-convex lens having radius of curvature R placed on plane glass plate as shown in fig. The circular interference fringes called Newton rings are formed when monochromatic light of wavelength is illuminated on



lens
on
The
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atic
on
ings
film
and



Plano convex lens. The radii of rings depend upon variable thickness of air film enclosed between plano convex lens and plane glass plate.



MAXIMUM INTENSITY

The condition of maximum intensity in Newton rings is

$$2nd = m\lambda$$

The refractive index of air is $n = 1$



$$d = \left(m + \frac{1}{2}\right) \frac{\lambda}{2} \quad \text{----- (1)}$$

Where $m = 0, 1, 2, 3, 4$ -----

From fig. we can write as

$$d = R - \sqrt{R^2 - r^2}$$

$$d = R - (R^2 - r^2)^{1/2}$$

Where r is the radius of Newton rings and $\frac{r}{R} \ll 1$

$$d = R - R \left(1 - \frac{r^2}{R^2}\right)^{1/2}$$

Expand above relation by Binomial theorem

$$d = R - R \left[1 - \frac{1}{2} \left(\frac{r}{R}\right)^2 + \text{-----}\right]$$

$$d = R - R + \frac{r^2}{2R} + \text{-----}$$

$$d = \frac{r^2}{2R} \quad \text{----- (2)}$$

Put eq (2) in eq (1)

$$\left(\frac{r^2}{2R}\right) = \left(m + \frac{1}{2}\right) \frac{\lambda}{2}$$

$$r_b = \sqrt{\left(m + \frac{1}{2}\right) \lambda R}$$

Put eq (2) in eq (1)

r_b

This is called radii of bright rings,

$$r_b = \sqrt{\frac{1}{2} \lambda R}$$

[For $m = 0$]

MINIMUM INTENSITY

and eq (3)

$$\frac{r^2}{2R} = \frac{m\lambda}{2}$$

$$r_d = \sqrt{m\lambda R}$$

$$m = 0, 1, 2, 3, \dots$$

dark fringes

$$r_d = \sqrt{m\lambda R}$$

$$2nd = m\lambda \quad \text{nil} \quad \angle a =$$

ml [For air n = 1]

$$d = \frac{\lambda}{2} \quad \dots (3)$$

Compare eq (2)

When

This is radius of dark fringes

r_d

$$r_d = 0 \quad [\text{Form} =$$

It simply shows $r_d > 0$ and $r_d = 0$ for $m = 0$. Hence central spot of Newton rings is dark because $r_d = 0$. It is experimental evidence of phase change of 180° on reflection of light from denser medium.

wave from denser

Q.8 What is Michelson Interferometer? Describe its principle, construction

and working. (PU. 2004, 2007, 2011, 2012, sU. 2013)

MICHELSON INTERFEROMETER

The instrument used to measure lengths or changes in length with great accuracy by means of interference fringes is called Michelson Interferometer.

PRINCIPLE

The Michelson Interferometer is based on interference of light. A beam splitter divides the monochromatic light into two parts which recombine and produces interference pattern.

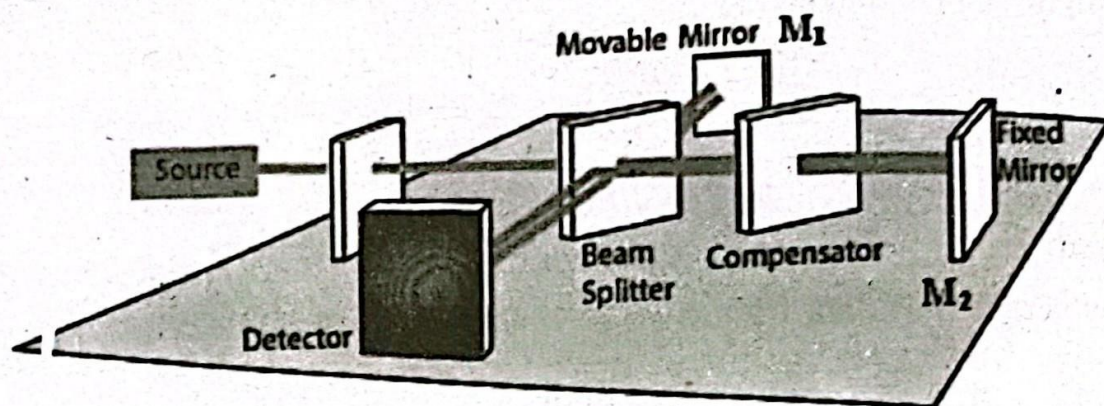
CONSTRUCTION

The essential parts of Michelson Interferometer are

The essential parts of Michelson Interferometer are

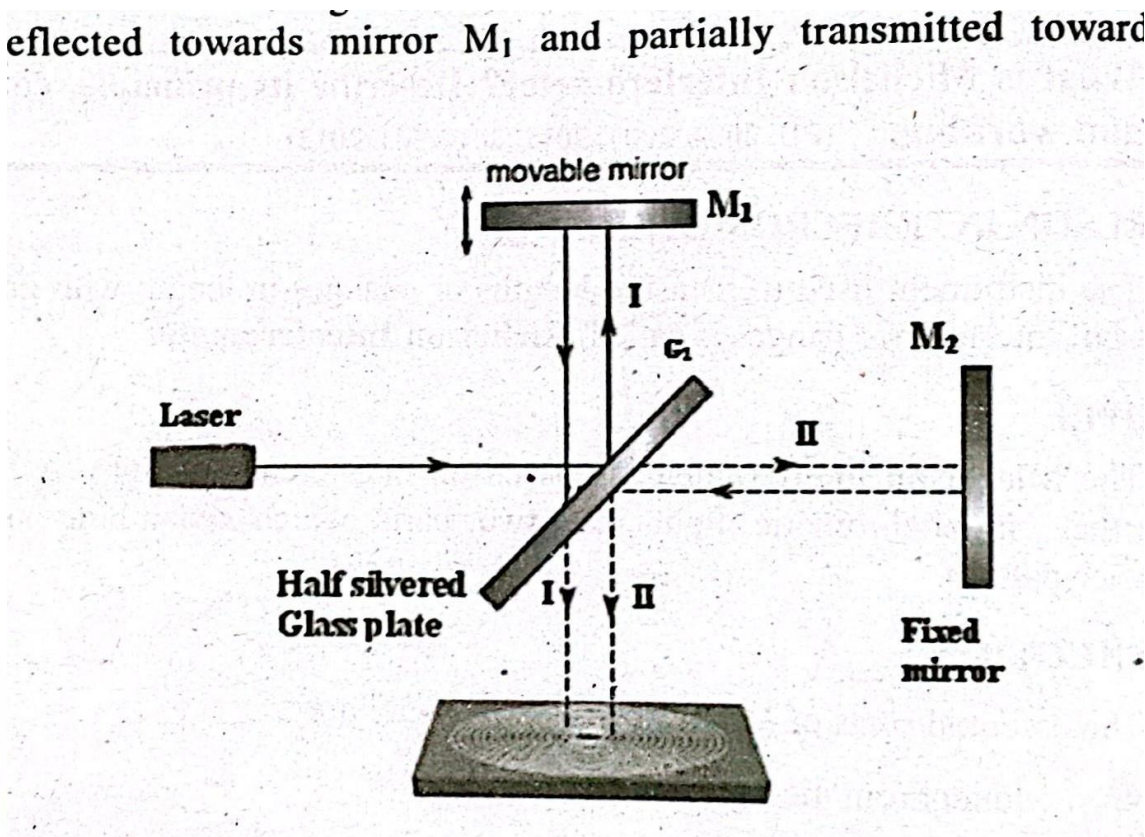
1. Monochromatic light source S
2. Movable mirror M_1
3. Fixed mirror M_2
4. Half silvered glass plate or beam splitter
5. Compensator

Detector



RAY DIAGRAM

The ray diagram of Michelson Interferometer is shown in fig. The light is reflected towards mirror M_1 and partially transmitted towards



monochromatic light from source S falls at angle of 45° on a half silvered glass plate (G). The light is partially reflected towards mirror M_1 and partially transmitted towards fixed mirror M_2 .

REFLECTION OF RAY 1.

The reflected portion called ray I covers distance L_1 and reaches M_1 which is reflected back towards glass plate G_1 . The half silvered glass plate G_1 partially transmits this portion which finally arrives at the detector.

REFLECTION OF RAY 11

The transmitted portion called ray II covers distance $1-2$ towards M_1 . The mirror M_2 reflects the beam back towards G_1 . The ray II partially reflected by G_1 also arrives at the detector.

FUNCTION OF GLASS PLATE G_2

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The glass plate G_2 which is cut from the piece of glass plate G_1 can be introduced in the path of ray 11 as a compensator plate. It can equalize the path length of the ray I and ray II in glass.

INTERFERENCE PATTERN

The ray I and ray II are phase coherent but having phase difference when combine. The interference pattern produced can be observed on detector by an observer. The observer sees a series of parallel interference fringes.

MOVEMENT OF M_1

The interference pattern changes when length 1.1 is changed by moving mirror M_1 with the help of a screw. A path difference $= \frac{1}{2} + \frac{1}{2}$ is produced for light ray I if M_1 is displaced through a distance equal to $X/2$. It means a bright fringe is replaced by next bright fringe or a dark fringe is replaced by next dark fringe whenever M_1 is moved by V_2 .

Similarly a bright fringe is replaced by next dark fringe or a dark fringe is replaced by next bright fringe whenever M_1 is moved through distance $X/4$ because a path difference $\frac{1}{2} = \frac{1}{4} + \frac{X}{4}$ is introduced for light ray I every time.

FRINGES

The distance L moved by mirror M_1 can be calculated by counting the number of dark fringes as

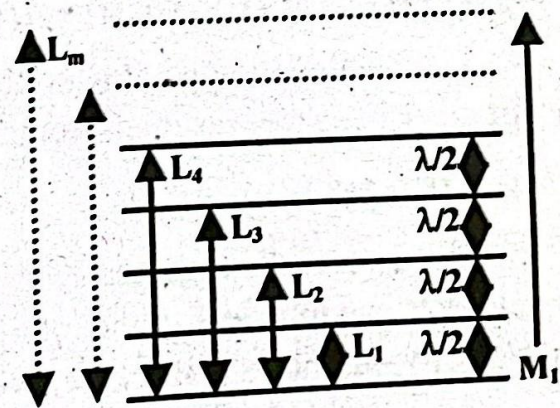
$$L_1 = \left(\frac{\lambda}{2}\right) = 1 \left(\frac{\lambda}{2}\right)$$

$$L_2 = \frac{\lambda}{2} + \frac{\lambda}{2} = 2 \left(\frac{\lambda}{2}\right)$$

$$L_3 = \frac{\lambda}{2} + \frac{\lambda}{2} + \frac{\lambda}{2} = 3 \left(\frac{\lambda}{2}\right)$$

$$L_4 = \frac{\lambda}{2} + \frac{\lambda}{2} + \frac{\lambda}{2} + \frac{\lambda}{2} = 4 \left(\frac{\lambda}{2}\right)$$

$$L_m = \frac{\lambda}{2} + \frac{\lambda}{2} + \frac{\lambda}{2} + \dots = m \frac{\lambda}{2}$$



COUNTING OF

bright fringes or dark

The formula of fringe counting is

The Michelson Interferometer is used to measure changes in length by counting number of interference fringes that pass the field of view as MI is moved.

Michelson measured the length of standard meter in terms of wavelength of a certain monochromatic red light emitted from a light source containing cadmium. The standard meter was equivalent to 1553163.5 wavelengths of red cadmium light and placed in Paris.

Q.9 Describe Michelson-Morley experiment, which proves ether does not exist in space? (PU. 2002, 2008, GCUF.2014)

ETHER HYPOTHESIS

The ether is a space filling substance or medium thought to be necessary for the propagation of light. It is supposed in 19th century that universe is filled with a medium such as ether for the propagation of light just as sound needs medium such as air.

OBJECT OF MICHELSON MORLEY EXPERIMENT

The Michelson and Morley experiment* was performed in 1881 about direct verification of existence of ether in space. The object of this experiment was to measure speed with which the earth moves through the ether.

APPARATUS OF MICHELSON MORLEY EXPERIMENT

The main apparatus used by Michelson and Morley for verification of existence of ether in space was interferometer.

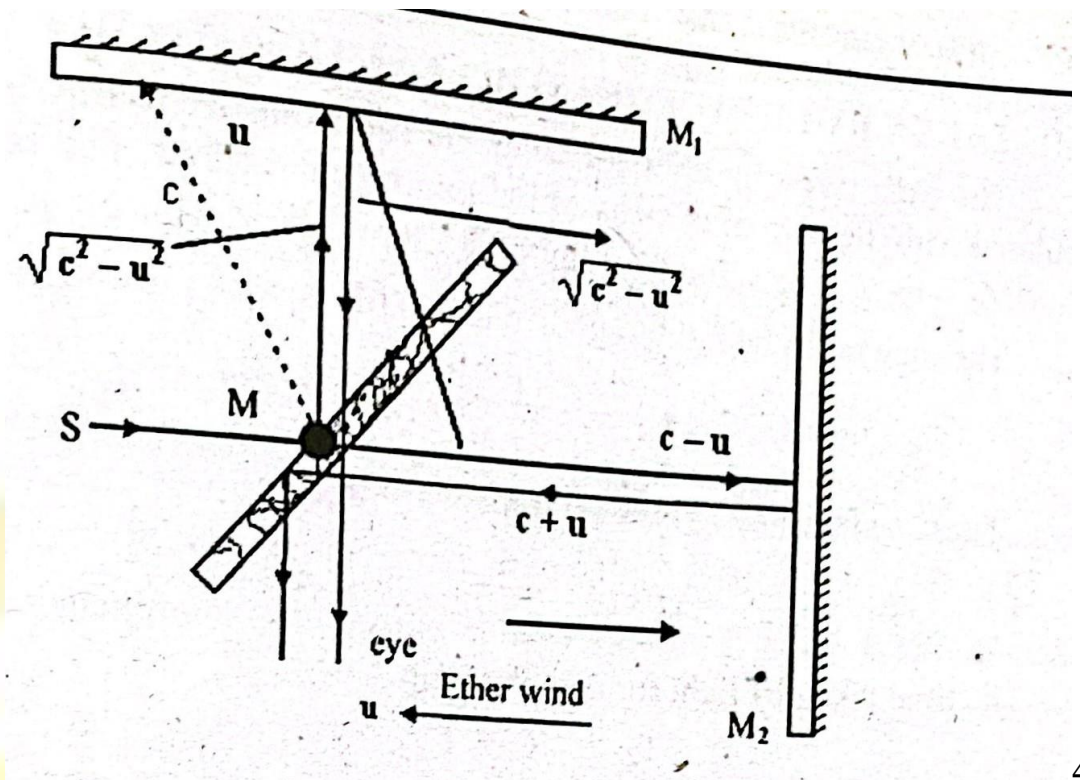
ETHER WIND

The earth together with interferometer is moving with velocity u through the ether. It is equivalent to the interferometer at rest with the ether streaming through it with the velocity $-u$. This is called ether wind.

WORKING

Consider an interferometer and earth together are moving with velocity u with respect to ether. The motion of interferometer gives rise to ether wind blowing in the direction of MS.

The ether wind will lower the velocity of light to $c - u$ in the direction MM_2 and increases the velocity to $c + u$ in the direction M_2M because light waves are carried by Ether, distance between M and M_1 and distance between M and M_2 is d .



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TIME TAKEN BY LIGHT FOR COMPLETE TRIP(M to M_1 & back to M)

The time taken by light for complete trip from M to M_1 & back to M can be calculated with the help of velocity of light in these directions.

The velocity of light along path MM_1 is $c - u$. The velocity of light from return path M_1M is also same $c + u$.

The time taken by light to go from M to M_1 is

$$\frac{d}{c - u}$$

The time taken by light to go from M_1 to M is

from M_1 to M is

$$\frac{d}{c + u}$$

The time taken by light for

complete trip

$$T_1 = \frac{d}{\sqrt{c^2 - u^2}} + \frac{d}{\sqrt{c^2 - u^2}}$$

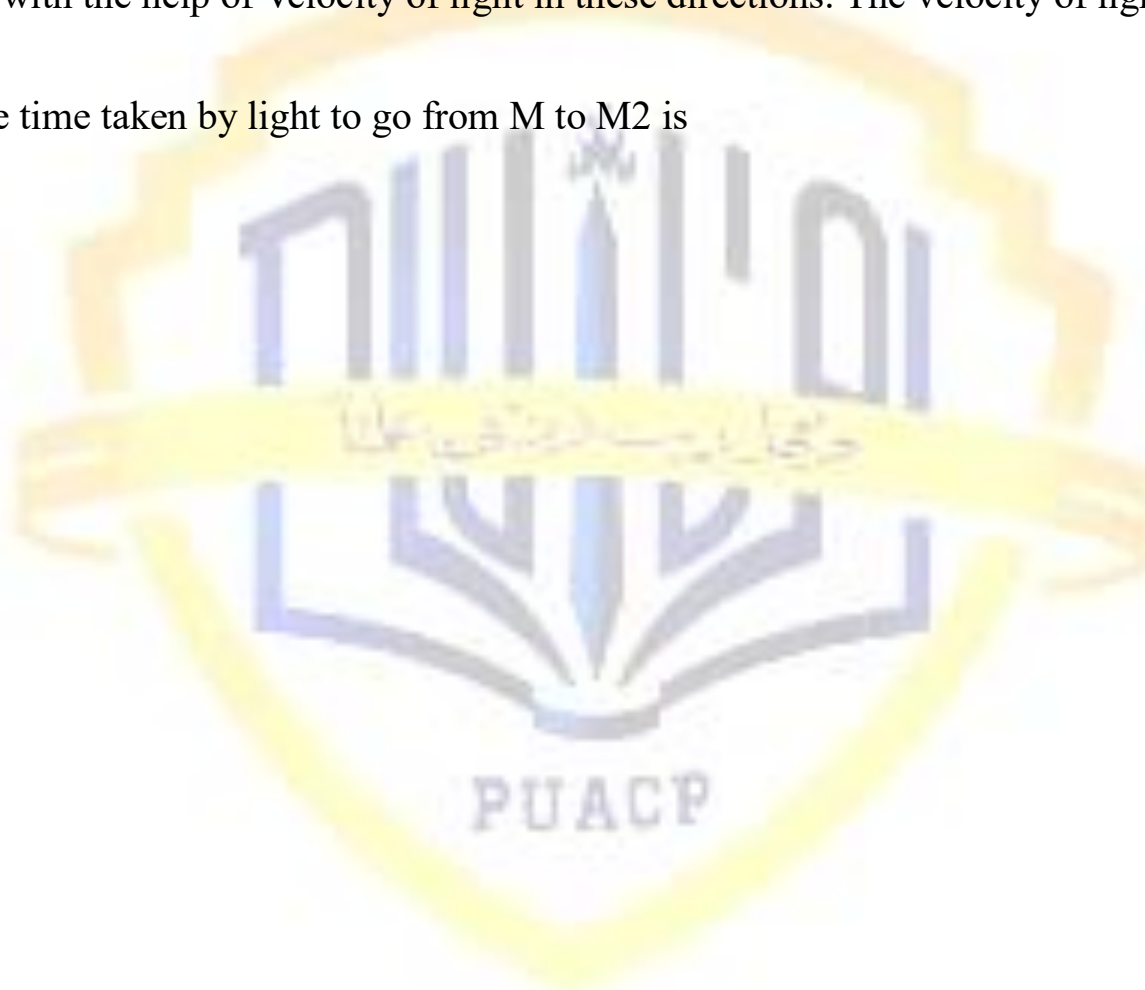
$$T_1 = \frac{2d}{(c^2 - u^2)^{1/2}} = \frac{2d}{c} \left(1 - \frac{u^2}{c^2}\right)^{-1/2}$$

$$= \frac{2d}{c} \left(1 + \frac{u^2}{c^2}\right) \quad [\text{Use Binomial theorem}]$$

TIME TAKEN BY LIGHT FOR COMPLETE TRIP(M to M2 & back to M)

The time taken by light for complete trip from M to M2 & back to M calculated with the help of Velocity of light in these directions. The velocity of light along

The time taken by light to go from M to M2 is



The time taken by light to go from M2 to M is

$$t_2 = \frac{d}{c+u}$$

The time taken by light for complete trip

$$T_2 = t_1 + t_2$$

$$T_2 = \frac{d}{c-u} + \frac{d}{c+u}$$

$$= \frac{d(c+u) + d(c-u)}{(c-u)(c+u)}$$

$$T_2 = \frac{2dc}{c^2 - u^2} = \frac{2dc}{c^2 (1 - u^2/c^2)}$$

$$T_2 = \frac{2d}{c} \left(1 - \frac{u^2}{c^2}\right)^{-1}$$

$$T_2 = \frac{2d}{c} \left(1 + \frac{u^2}{c^2}\right) \quad [\text{Use Binomial theorem}]$$

The difference of time interval between two trips is

$$\Delta t = T_2 - T_1$$

$$\Delta t = \frac{2d}{c} \left(1 + \frac{u^2}{c^2}\right) - \frac{2d}{c} \left(1 - \frac{u^2}{2c^2}\right)$$

$$\Delta t = \frac{2d}{c} \left(1 + \frac{u^2}{c^2} - 1 + \frac{u^2}{2c^2}\right)$$

$$= \frac{2d}{c} \left(1 + \frac{1}{2}\right) \frac{u^2}{c^2}$$

$$\Delta t = \frac{2d}{c} \left(\frac{u^2}{2c^2}\right) = \frac{du^2}{c^3}$$

The path difference of two light waves related to time difference is

On comparing

$$n = \frac{nc \Delta t}{d} = \frac{2d}{c^2}$$

The path difference corresponding to shifting of fringes is

480 [M.B.DJ WAVES & OSCILLATI = $n\lambda$

Now rotate the interferometer through 90° . This interchange the roles of the two light paths. The time difference between the two waves entering the eye is also reversed.

This change the phase difference between the combining waves and alters the position of interference maxima. The change in time difference is $2\Delta t$.

The expected maximum shift in the number of fringes when interferometer is rotated through

$$\Delta n = \frac{c \cdot 2\Delta t}{\lambda} = \frac{2u^2 d}{\lambda c^2}$$

90° is

The distance $d = 11$ m in Michelson—Morley interferometer obtained by multiple reflections and $= 5.9 \times 10^{-7}$ m. The ratio $u = 10^4$ if u is assumed roughly the orbital speed of earth. The numerical value of expected maximum shift in the number of fringes when interferometer

rotated through 90° is

$$\Delta n = \frac{2(11)}{5.9 \times 10^{-7}} (10^{-4})^2$$

$$\Delta n = 0.4$$

is rotated

Michelson and Morley were confident that they could observe a shift of 0.01 in number of fringes when extra care is taken in experiment. Hence there is no observable shift in fringes.

CONCLUSION

The conclusion of Michelson and Morley experiment is

① There ether does not exist in space because there will fringe shift if ether is present in space.

② The speed of light in free space is the same everywhere and its value does not depend upon velocity of source or observer.

SAMPLE PROBLEMS

Resnick, Halliday, Krane

is illuminated with light from mercury vapor lamp. (Y, Krane)

P.] The double slit arrangement is illuminated with light from mercury vapor lamp filtered so that only the strong green line ($\lambda = 546\text{nm}$) is visible. The slits are 0.12 mm apart, and the screen on which the interference pattern is appeared is 55 cm away. What is the angular position of first minimum? of the tenth minimum. What is the linear distance on screen between the adjacent maxima m and $m + 1$? (PU. 2003, 2006, 2009, SU. 2011, 2012). Sol.

DATA

$$\lambda = 546\text{ nm} = 546 \times 10^{-9}\text{ m}$$

$$d = 0.12\text{ mm} = 0.12 \times 10^{-3}\text{ m}$$

$$L = 55\text{ cm} = 55 \times 10^{-2}\text{ m}$$

$$\theta_1 = ?$$

$$\theta_{10} = ?$$

$$\Delta y = ?$$

CALCULATIONS

For first minimum, $m = 0$

$$d \sin \theta_1 = (m + \frac{1}{2})\lambda$$

$$0.12 \times 10^{-3} \sin \theta_1 = (0 + \frac{1}{2}) 546 \times 10^{-9}$$

$$\theta_1 = \sin^{-1}(0.00227) =$$

O. 130.

For tenth minimum, $m = 9$

$$d \sin \theta_{10} = (m + \frac{1}{2})\lambda$$

$$0.12 \times 10^{-3} \sin \theta_{10} = (9 + \frac{1}{2}) 546 \times 10^{-9}$$

$$\theta_{10} = \sin^{-1}(0.0477) = 2.70^\circ$$

LINEAR DISTANCE

$$\Delta y = \frac{\lambda L}{d}$$

$$\Delta y = \frac{546 \times 10^{-9} \times 55 \times 10^{-2}}{0.12 \times 10^{-3}}$$

$$0.12 \times 10^{-3}$$

=

$$2.50 \times 10^{-3} \text{ m}$$

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P.2 Find the resultant of the following wave disturbances
 $E_1 = E_0 \sin \omega t$, $E_2 = E_0 \sin(\omega t + 15^\circ)$, $E_3 = E_0 \sin(\omega t + 30^\circ)$,
 $E_4 = E_0 \sin(\omega t + 45^\circ)$.

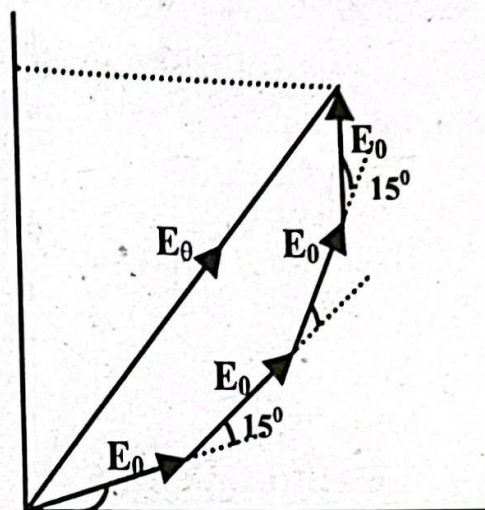
Sol.

The phasor diagram is shown in fig. The phasor angle ϕ is 15° .

The amplitude E_0 is 3.8 times as long as E_0 and resultant phasor makes angle 22.5° with to E_1 as measured with ruler and protractor.

$$E = E_1 + E_2 + E_3 + E_4$$

$$E = 3.8 E_0 \sin(\omega t + 22.5^\circ)$$



Find graphically the resultant of the following wave disturbances

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$$\sin \omega t, E_2 = E_0 \sin(\omega t + 150^\circ), E_3 = E_0 \sin(\omega t + 300^\circ),$$

phasor

PUACP

$$\sin(\theta + 22.5)$$

air
it is

A water film with $n = 1.33$ is 320 nm thick. If illuminated with white light at normal incidence. What color will it appear to be in reflected light.

p.3 1.33 in
is 320 nm thick. If
illuminated with

DATA

$$\begin{aligned} n &= 1.33 \\ d &= 320 \text{ nm} \\ \lambda &= ? \end{aligned}$$

CALCULATIONS

For maxima

$$\begin{aligned} 2nd &= (m + \frac{1}{2}) \lambda \\ \lambda &= \frac{2nd}{(m + \frac{1}{2})} = \frac{2 \times 1.33 \times 320}{(m + \frac{1}{2})} = \frac{851}{(m + \frac{1}{2})} \end{aligned}$$

What color will it appear to be in reflected light.

Sol.

DATA

$$\underline{851 \text{ nm}}$$

+ 1/2)

For minima

$$\begin{aligned} 2nd &= m \lambda \\ \lambda &= \frac{2 \times 1.33 \times 320}{m} = \frac{851 \text{ nm}}{m} \end{aligned}$$

Maxima and minima occurs at following wavelengths

m	0(max)	1(min)	1(max)	2(min)	2(max)
$\lambda(\text{nm})$	1702	851	567	426	340

to $m = 1$ lies in visible region (400 nm to 700 nm).

Only maxima corresponding to $m = 0$

The light of wavelength 567 nm appears yellow green.

Resnick, Halliday,

Lenses are often coated with thin films of transparent substances such as MgF_2 with $n = 1.38$ to reduce the reflection from the glass surface. **How**

thick a coating is needed to produce a minimum reflection at the center of the visible spectrum ($\lambda = 550 \text{ nm}$) Sol.

DATA

$$n = 1.38$$

$$\lambda = 550 \text{ nm}$$

$$d = ?$$

CALCULATIONS

$$2nd = (m + \frac{1}{2})\lambda$$

$$d = \frac{(m + \frac{1}{2})\lambda}{2n} = \frac{(0 + \frac{1}{2})\lambda}{2(1.38)}$$

$$d = 100 \text{ nm}$$

PS Yellow light of wavelength 589 nm illuminates a Michelson interferometer. How many bright fringes will be counted if mirror is moved by 1.0 cm? (PU. 2011, 2012)

Sol.

DATA

$$L = 1.0 \text{ cm} = 1.0 \times 10^{-2} \text{ m}$$

$$\lambda = 589 \text{ nm} = 589 \times 10^{-9} \text{ m}$$

$$m = ?$$

CALCULATIONS

$$\frac{589}{1.0 \times 10^9}$$

$$1.0 \times$$

$$m \times \frac{\lambda}{2}$$

$$m = 3$$

Monochromatic green light wavelength 554 nm, illuminates parallel narrow slits 7.7 mm apart. Calculate the angular deviation of the third order bright fringe in radians and in degrees.

DATA

$$m = 3$$

$$554\text{nm} = 554 \times 10^{-9} \text{ m}$$

$$7.7 = 10^{-6} \text{ m}$$

$$03$$

CALCULATIONS

$$d \sin \theta$$

$$7.7 \times 10^{-6} \sin \theta = 3 \times 554 \times 10^{-9}$$

$$\theta = \sin^{-1} (0.215) = 124$$

$$\theta = 0.215 \text{ rad} \quad [\text{Use } 180^\circ = \pi \text{ rad}]$$

P.2. In double slit experiment to demonstrate the interference of light the separation of narrow slits is made doubled, In order to maintain the same spacing of the fringe on the screen, how much the distance D or L be altered. The wavelength of light remains unchanged,

Sol.

Length of light = λ
 Separation of narrow slits = d
 Fringe spacing or fringe width = Δy
 Distance between screen and slits = L
 New separation of slits = $d' = 2d$
 New distance between screen & slits = $L' = ?$

DATA

Wavelength of light

Separation of narrow slits

Fringe

spacing or fringe width

CALCULATIONS

$$L = \frac{\Delta y d}{\lambda}$$

$$L' = \frac{\Delta y d'}{\lambda} = \frac{\Delta y (2d)}{\lambda} = 2L$$

CALCULATIONS

For

Resnick, Halliday,

P.3 A double slit experiment is performed with blue green light of wavelength 512 nm. The slits are 1.2 mm apart and screen is 5.4 m from the slits. How far apart are the bright fringes as seen on the screen.

(Pu. 2011, 2012, 2013, su. 2012, 2013)

Sol.

DATA

For bright fringes

$$\lambda = 512 \text{ nm} = 512 \times 10^{-9} \text{ m} \quad d = 1.2$$

$$\text{mm} = 1.2 \times 10^{-3} \text{ m}$$

$$L = 5.4 \text{ m}$$

$$\Delta y = ?$$

CALCULATIONS

$$\Delta y = \frac{\lambda L}{d}$$

$$\Delta y = \frac{512 \times 10^{-9} \times 5.4}{1.2 \times 10^{-3}}$$

$$= 2.30 \times 10^{-3} \text{ m}$$

P.4

Find the slit separation of a double slit experiment that will produce bright interference fringes 1° apart in angular separation. Assume a wavelength of light is 592 nm.

Sol.

DATA

$$\lambda = 592 \text{ nm} = 592 \times 10^{-9} \text{ m}$$

$$\Delta\theta = 1^\circ =$$

$$0.0174 \text{ rad}$$

d

$$= ?$$

CALCULATIONS

$$\Delta\theta = \frac{\lambda}{d}$$

$$0.0174 = \frac{592 \times 10^{-9}}{d}$$

$$d = 3.40 \times 10^{-5} \text{ m}$$

$$3.40 \times 10^{-5} \text{ m}$$

$$\lambda = 589 \text{ nm} = 589 \times 10^{-9} \text{ m}$$

$$\Delta\theta = 0.23^\circ = 4.012 \times 10^{-3} \text{ rad}$$

$$4.44 \times 10^{-3} \text{ rad}$$

$$\Delta\theta' = \Delta\theta + 10\% \text{ of } \Delta\theta$$

$$\Delta\theta' = 4.012 \times 10^{-3} + 10\% \times 4.012 \times 10^{-3} =$$

$$\lambda' = ?$$

CALCULATIONS

$$\Delta\theta = \frac{\lambda}{d}$$

$$4.012 \times 10^{-3} = \frac{589 \times 10^{-9}}{d}$$

$$d = 1.46 \times 10^{-4} \text{ m}$$

Apply the eq

$$\lambda' = \Delta\theta' \times d$$

$$\lambda' = 4.44 \times 10^{-3} \times 1.46 \times 10^{-4}$$

$$\lambda' = 6.4 \times 10^{-7} \text{ m}$$

Now apply

A double slit arrangement produces interference fringes for sodium light of wavelength 589 nm that are 0.23° apart. For what wavelength would the angular separation be 10 % greater? Assume that θ is small.

DATA

A double slit arrangement produces interference fringes for sodium light of wavelength 589 nm that are 0.20° apart. What is the angular separation if

entire arrangement is immersed in water ($n = 1.33$).

DATA

$$\begin{aligned}\lambda &= 589 \text{ nm} = 589 \times 10^{-9} \text{ m} \\ \Delta\theta &= 0.20^\circ = 3.48 \times 10^{-3} \text{ rad} \\ n &= 1.33 \\ \Delta\theta' &= ?\end{aligned}$$

$$\lambda_n = \frac{\lambda}{n}$$

CALCULATIONS

$$\Delta\theta = \frac{\lambda}{d}$$

er use λ_n

$$\Delta\theta' = \frac{\lambda_n}{d} = \frac{1}{d} \left(\frac{\lambda}{n} \right) \Delta\theta' = \frac{\Delta\theta}{n}$$

For water use

Resnick, Halliday,

P.7

$$\Delta\theta' = \frac{3.48 \times 10^{-3}}{1.33} = 2.62 \times 10^{-3} \text{ rad}$$

ima of a c

If the distance between the first and tenth minima of a double slit pattern is 18 mm and the slits are separated by 0.15 mm with the screen 50 cm from the slits. What is the wavelength of light used?

sol.

DATA

$$y_{10-1} = 18 \text{ mm} = 18 \times 10^{-3} \text{ m} \quad d = 0.15 \text{ mm} = 0.15 \times 10^{-3} \text{ m}$$

$$50 \text{ cm} = 50 \times 10^{-2} \text{ m}$$

CALCULATIONS

For dark or minima

$$y_m = (m + \frac{1}{2}) \frac{\lambda L}{d}$$

Order of dark fringe is one, for $m = 0$ and order is 10 for $m = 9$

$$y_1 = (0 + \frac{1}{2}) \frac{\lambda L}{d}$$

$$y_{10} = (9 + \frac{1}{2}) \frac{\lambda L}{d}$$

$$y_{10} - y_1 = (9 + \frac{1}{2}) \frac{\lambda L}{d} - (0 + \frac{1}{2}) \frac{\lambda L}{d}$$

$$y_{10} - y_1 = \frac{9\lambda L}{d} + \frac{\lambda L}{2d} - \frac{\lambda L}{2d} = \frac{9\lambda L}{d}$$

$$18 \times 10^{-3} = 9 \times \frac{\lambda \times 50 \times 10^{-2}}{0.15 \times 10^{-3}}$$

$$\lambda = 6 \times 10^{-7} \text{ m}$$

For dark or

P.8 If a mirror M in interferometer is moved through 0.233 mm, 792 fringes are counted with a light meter. What is the wavelength of the light?

(PU. 2002, 2008, GCUF. 2014) sol.

DATA

$$L = 0.233 \text{ mm} = 0.233 \times 10^{-3} \text{ m}$$

$$m = 792$$

$$\lambda = 5.88 \times 10^{-7} \text{ m}$$

CALCULATIONS

$$L =$$

$$0.233 \times 10^{-3}$$

$$792 \times$$